MILITARY SPECIFICATION SHEET

GENERATOR SET, GASOLINE ENGINE DRIVEN,

3 KW, 400 HERTZ

TYPE I (TACTICAL), CLASS 2 (UTILITY)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator set described herein shall consist of this document and the issue in effect of specification MIL-G-52732.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for mode II, size 3 sets, model MEP-021A.

1.2 Classification.

1.2.1 Set rating. The mode II set shall be rated 0.8 power factor (PF) lagging and shall be connectable for the following voltage ratings:

(a) 1 phase, 2 wire, 120 volts.
(b) 1 phase, 2 wire, 240 volts.
(c) 3 phase, 3 wire, 120 volts line-to-line.
(d) 3 phase, 4 wire, 120 volts line-to-neutral and 208 volts line-to-line (120/208 volts).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-H-52112	- Heating Torches, Equipment Winterization, Instant Lighting.
MIL-G-52732	- Generator Sets, Gasoline Engine Driven, 0.5 KW thru 10 KW, 60 Hertz and 28 Volt Direct Current, Type I (Tactical), Class 2 (Utility) General Specification for.

FSC 6115

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STANDARDS

Military

MS39299	- Engine, Gasoline, Air-Cooled, Industrial-Type, 4 Cycle, 6 Net Continuous HP.

DRAWINGS

ME

TA13213E4098	- Generator Set, Gasoline Engine Driven, 3 KW, AC, 400 Hz, Air-Cooled, Portable Tubular Frame, Skid Mounted.
TA13213E4430	- Winterization Kit, 3 KW, Generator Sets.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements shall be in accordance with MIL-G-52732 and as specified herein.

3.2 Description. The set shall be in accordance with Top Assembly TA13213E4095 and as specified herein.

3.3 Starting and operating. Servicing, adjustment and replacement shall be as specified in table I.

3.3.1 With winterization. At ambient temperatures from minus 25° F. (minus 31.7° C.) to minus 65° F. (minus 53.9° C.), the set shall start within 15 minutes with the application of external heat from a torch conforming to MIL-H-52112, type I. The set shall meet these requirements utilizing a winterization kit in accordance with TA13213E4430.

3.4 Maximum power. The maximum power (corrected) in kilowatts of the set shall be not less than 4.5 kw when operated at rated speed, voltage, and at unity power factor.

3.5 Engine.

3.5.1 Engine design. The Government furnished engine will conform to MS39299.

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3.5.2 Engine temperature. Engine temperatures shall not exceed the following:

Location	Maximum temperatures	
	First 10 hours of engine operation	After 10 hours of engine operation
Sump oil	300° F. (148.9° C.)	265° F. (129.2° C.)
Cylinder head	500° F. (260° C.)	475° F. (246.1° C.)
Exhaust	1200° F. (648.9° C.)	1200° F. (648.9° C.)

Table I. Schedule of Servicing and Adjusting

All Items May Be Serviced Before Test

Item No.	Hours between service periods	Item
1	8	Any minor maintenance such as retightening and retorquing nuts, bolts and screws that can be performed without disturbing or removing adjacent components, accessories, or shrouding (except top shroud, pump, intake manifold and exhaust system).
2	Not allowed	Any major maintenance that requires removal of other components, accessories or shrouding such as oil pans, cylinder heads and connecting rods.
3	As required	Cleaning and regapping or replacing spark plugs.
4	As required	Servicing fuel filter.
5	100	Checking compression pressures.
6	8	Checking and adding lubricating oil, as necessary.

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Table I. Schedule of Servicing and Adjusting (continued)

All Items May be Serviced Before Test

Item No.	Hours between service periods	Item
7	8	Adjusting carburetor.
8	As required	Cleaning carburetor.
9	8	Adjusting governor (normal speed adjustment).
10	100	Changing lubricating oil.
11	100	Changing lubricating oil filter element.
12	100	Servicing air cleaner.
13	300	Adjusting and dressing breaker points.
14	300	Adjusting ignition timing.
15	250	Valve adjustment.
16	1500	Cleaning and inspection of combustion chamber, manifolds and valves.
17	500	Replacing breaker points.
18	As required	Replacing condenser.
19	*	Gasket replacement.

* Not allowed except where component is removed for servicing.

3.5.3 Governing system.

3.5.3.1 Short term, steady-state stability (30 seconds), mode II and III sets. The short term steady-state stability bandwidth percentage shall not exceed 1 percent.

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3.5.3.2 Long term, steady-state stability (4 hours), mode II and III sets. The long term steady-state stability band width percentage shall not exceed 2 percent.

3.6 Generator.

3.6.1 Voltage unbalance, mode II and III sets. The variation between line-to-line voltages shall not exceed 5 percent.

3.6.2 Phase balance (voltage), mode II and III sets. The difference in line-to-line and line-to-neutral voltages shall not exceed 1 percent.

3.6.3 Voltage waveform, mode II and III sets. The deviation factor of the line-to-neutral and line-to-line voltages for 3-phase generator connections at rated load, no-load and at unity power factor rated kw shall not exceed 5.0 percent. Under the same conditions, the deviation factor at the single phase connections shall not exceed 6.0 percent. Single frequency harmonics shall not exceed 3.0 percent.

3.6.4 Output voltage ripple, mode IV sets. Not applicable for the mode II, size 3 set.

3.6.5 Saturation curves. With the generator temperature stabilized at normal ambient conditions the exciter field current for rated voltage shall not exceed 0.60 ampere at no-load and 0.95 ampere at rated load.

3.6.6 Voltage adjustment range. The minimum acceptable voltage adjustment range shall be 5 percent above and below the rated voltage.

3.6.7 Voltage regulation. The voltage regulation percentage shall not exceed 4 percent.

3.6.8 Voltage steady-state performance. The voltage steady-state stability bandwidth percentage shall not exceed 2 percent.

3.6.9 Voltage transient performance. The voltage dip or rise percentage shall not exceed 30 percent.

3.6.10 Generator efficiency. The generator efficiency shall not be less than 80 percent at 1.0 PF and 75 percent at 0.8 PF.

3.7 Circuit interrupter. The circuit interrupter shall open within 0.25 seconds under the condition of a short circuit in any one phase or in all phases simultaneously.

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3.8 Instruments. The inaccuracy or error of the metering system shall not exceed the following:

Instrument	Allowable Value
Ammeter (percent load)	10 percent inaccuracy
Voltmeter	3 percent inaccuracy
Frequency meter	5 Hz error

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. The quality assurance provisions shall be in accordance with MIL-G-52732 and as specified herein.

4.2 Engine preventative maintenance services. The engine preventative maintenance services shall be performed as specified in table I.

4.3 Drop test (free fall). The height of the drop shall be 18 inches for mode II, size 3 set.

4.4 Vibration test. The handrail or shock mount resonances shall not be utilized for resonant dwell conditions.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-G-52732.

6. NOTES

6.1 Notes. The notes specified in MIL-G-52732 are applicable to this specification and as specified herein.

6.2 Stable operation (voltage). Stable operation is operation within a bandwidth of 2 percent of the rated voltage.

6.3 International standardization. Not applicable for the mode II, size 3 set.

Custodians:
Army - ME
Air Force - 80
Navy - YD

Review activities:
Army - EL, MI, MU

Preparing activity:
Army - ME

Project No. 6115-0144-7

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-G-[illegible] - Generator Set, Gasoline Engine Driven, 3 KW, 400 Hertz Type I (Tactical), Class 2 (Utility)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there is any unit price cost anticipated, indicate it here and on a separate page.)

5. SUBMITTED BY (Printed or typed name and activity) (Optional)

DD FORM 1426

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